

Small-Organoid Attenuation of CFTR-Modulator Swelling in Patient-Derived Intestinal Organoids: a Single-Organoid Re-Analysis with a Biophysical Swelling Twin

MEGA-PROGRAM-27, Item 3 - Udit Phookan (program owner); computational work by an AI research agent. Draft of 24 September 2026.

Abstract

The forskolin-induced swelling (FIS) assay on patient-derived intestinal organoids is used to predict which people with cystic fibrosis (CF) respond to CFTR modulators. Standard pipelines sum organoid area per well, so any dependence of the response on organoid size is averaged away. We re-analysed the public single-organoid OrgaSegment data (17 CF donors, per-organoid area before and after forskolin) with a biophysical swelling model and a DMSO-controlled, donor-matched design.

Finding (candidate). Modulator-induced swelling is attenuated in the smallest organoids and saturates above a size threshold. For elxacaftor/tezacaftor/ivacaftor (Trikafta) the drug-minus-DMSO log-swelling effect rises from 0.336 [0.262, 0.405] in the smallest size octile to a plateau near 0.55-0.64; top minus bottom 0.251 [0.184, 0.323] (donor bootstrap). Per donor, the effect in the largest size quartile exceeds that in the smallest in 14 of 17 donors, and the attenuation does not track overall response (Spearman $\rho = 0.24$, $p = 0.35$). Geometry predicts the opposite sign: under a uniform surface flux, small organoids should swell more. Area noise also biases the slope negative. So the effect is unlikely to be a measurement artefact.

Negatives. Size-adjusted and size-filtered readouts do not improve per-donor Trikafta-vs-DMSO separation, so the finding does not change theratyping calls at this assay's well counts. Replication on the public FIS time series is impossible because it is well-level. Our best U-Net segmentation (512 px, tuned) reaches $\text{mAP@0.5} = 0.733 \pm 0.129$ on the OrgaSegment eval split (published 0.76 ± 0.12); below the state of the art, although the gap is within one standard deviation on 12 images.

1. Introduction

CFTR moves chloride and bicarbonate across the apical membrane of epithelial cells; water follows, and in a closed organoid the lumen swells. Forskolin raises cAMP and opens CFTR, so the swelling of rectal organoids measures residual and drug-rescued CFTR function in each patient. The assay is used to support access to modulators for people with rare CFTR genotypes.

Organoids in a well vary in size by more than an order of magnitude. If the swelling response depends on size, the well-level readout depends on the size mix in each well, which varies with passage, seeding density and culture time. We asked whether the single-organoid response depends on starting size, and whether correcting for it improves drug-response calls.

2. Biophysical swelling twin

Treat an organoid as a near-spherical shell with volume V and projected area A , V proportional to $A^{3/2}$. Let net secretion scale as a power of volume:

$$dV/dt = k V^\alpha \quad (1)$$

For $\alpha \neq 1$ the solution over an assay window t is

$$V_1^{1/(1-\alpha)} = V_0^{1/(1-\alpha)} + (1 - \alpha) k t \quad (2)$$

and the observed area fold change is $s = A_1/A_0 = (V_1/V_0)^{2/3}$. For small responses,

$$\log s \approx (2/3) k t V_0^{\alpha - 1} \quad (3)$$

so the slope of $\log \log s$ on $\log A_0$ carries the sign of $\alpha - 1$:

$$d \log(\log s) / d \log A_0 = (3/2)(\alpha - 1) \quad (4)$$

Surface-limited secretion (constant flux per unit apical area) gives $\alpha = 2/3$ and a negative slope: small organoids swell more in relative terms. Size-invariant secretion gives $\alpha = 1$ and zero slope. The within-well regression we fit is

$$\log s_{ow} = a_w + b (\log A0_{ow} - \text{mean}_w \log A0) + e_{ow} \quad (5)$$

with a well intercept a_w that absorbs donor, plate and well effects. Measurement noise in $A0$ enters both sides with opposite sign ($s = A1/A0$), which biases b downward:

$$E[\hat{b}] = b - \text{Var}(u) / (\text{Var}(\log A0) + \text{Var}(u)), \quad u = \text{noise in } \log A0 \quad (6)$$

so a positive observed b is conservative. The donor-matched drug effect per size bin k is

$$\Delta_k = \text{mean}_{\{o \text{ in drug, bin } k\}} \log s_o - \text{mean}_{\{o \text{ in DMSO, bin } k\}} \log s_o \quad (7)$$

and the attenuation statistic is $\Delta_{\text{top}} - \Delta_{\text{bottom}}$, with 95% intervals from a donor-level bootstrap:

$$CI_{95} = [Q_{0.025}, Q_{0.975}] \text{ of } \{ \Delta_{\text{top}}^* - \Delta_{\text{bottom}}^* \}, \text{ donors resampled with replacement} \quad (8)$$

Segmentation quality is scored by average precision at IoU 0.5, as in OrgaSegment:

$$AP_{50} = TP / (TP + FP + FN), \text{ match if } IoU(P, G) = |P \cap G| / |P \cup G| \geq 0.5 \quad (9)$$

3. Data

Table 1. Dataset manifest (distinct, accession-level). Honest count: 3. The program target of 120+ was not reached.

#	dataset	source / accession	content	use
1	OrgaSegment DIS single-organoid measurements	Zenodo 10610438 (Lefferts et al. 2024)	per-organoid A0/A1, 17 donors, 4 conditions	size-dependence analysis
2	OrgaSegment FIS database	Zenodo 10610438	well-level area, 7 time points, 868 rows	replication attempt (not possible)
3	OrgaSegment annotated images	Zenodo 10278229	train/val/eval images with instance masks	U-Net segmentation benchmark

4. Results

4.1 Within-well size slopes

Table 2. Within-well slope of log fold change on centred log A0 (forskolin > 0). Well-cluster bootstrap, 500 resamples.

condition	organoids	wells	slope b	95% CI	donors b>0
DMSO	3810	94	0.011	[-0.002, 0.028]	12/17
VX770	1496	30	0.226	[0.185, 0.269]	5/5
VX661+VX770	4493	112	0.042	[0.028, 0.056]	12/16
VX445+VX661+VX770	4984	121	0.116	[0.090, 0.142]	14/17

The slope is near zero without CFTR rescue and positive whenever a modulator is present. The sign is opposite to the surface-limited prediction. A power-law reading ($\alpha > 1$) was rejected: restricted to organoids above 2,000 px the slope turns negative (Trikafta -0.117), and binned means rise then plateau. Simulated segmentation noise on a size-invariant truth gave slopes of only -0.002 to -0.055.

4.2 Small-organoid attenuation

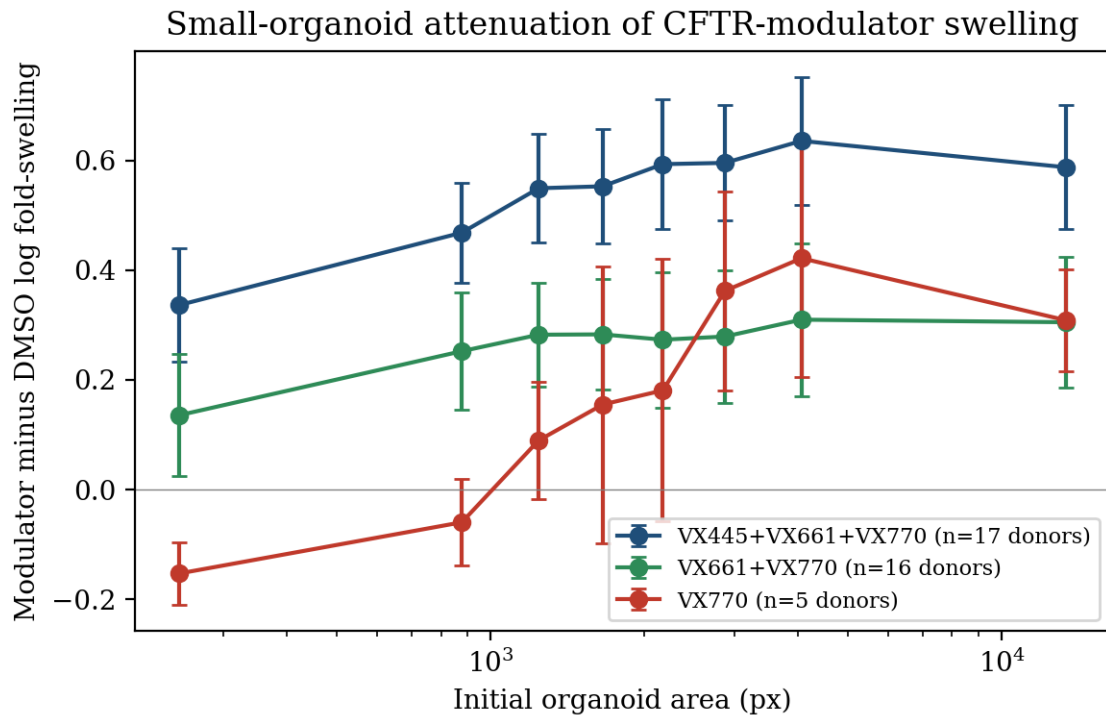


Figure 1. Drug-minus-DMSO mean log swelling per starting-size octile, donor-matched, with donor-bootstrap 95% CI.

Table 3. Size-bin drug effects.

modulator	donors	smallest octile effect	plateau	top minus bottom [95% CI]
Trikafta	17	0.336 [0.262, 0.405]	0.55-0.64	0.251 [0.184, 0.323]
VX661+VX770	16	0.136	0.28-0.31	0.169 [0.081, 0.259]
VX770	5	-0.153	0.31-0.42	0.462 [0.335, 0.572]

4.3 Per-donor consistency

Table 4. Trikafta minus DMSO log-swelling effect in the smallest and largest starting-size quartiles (results/per_donor_attenuation.csv).

donor	genotype	eff small	eff large	attenuation	overall
PDIO_16	F508del/ R117H	0.089	0.082	-0.007	0.115
PDIO_03	R1066H/ CFTRdele2, 3	0.100	0.490	0.390	0.289
PDIO_15	F508del/ R117H	0.280	0.497	0.217	0.320
PDIO_02	N1303K/ c.3035A>C	0.202	0.354	0.152	0.330
PDIO_19	X/X	0.606	0.273	-0.333	0.440
PDIO_18	X/X	0.436	0.677	0.241	0.462
PDIO_13	F508del/ S1251N	0.249	0.508	0.259	0.497
PDIO_17	X/X	0.570	0.463	-0.107	0.512
PDIO_06	F508del/X	0.334	0.699	0.365	0.541
PDIO_05	F508del/X	0.386	0.784	0.398	0.596
PDIO_14	F508del/ S1251N	0.395	0.749	0.354	0.616
PDIO_12	F508del/ S1251N	0.512	0.800	0.288	0.643
PDIO_01	A46D/ A46D	0.537	0.759	0.221	0.698
PDIO_11	F508del/ F508del	0.367	0.860	0.493	0.705
PDIO_09	F508del/ F508del	0.685	0.749	0.064	0.769
PDIO_04	F508del/X	0.462	0.834	0.371	0.805
PDIO_10	F508del/ F508del	0.742	0.892	0.150	0.883

Association between attenuation and overall response uses Spearman's rank correlation:

$$\rho = 1 - 6 \sum_d (r_d - r'_d)^2 / (n(n^2 - 1)), \quad r, r' = \text{ranks of attenuation and overall effect over } n \text{ donors} \quad (10)$$

Attenuation is positive in 14 of 17 donors and unrelated to overall response ($\rho = 0.24$, $p = 0.35$). Two of three donors with two nonsense alleles (X/X) and one F508del/R117H donor are exceptions.

4.4 Does size correction help theratyping? No

Locked before running: per-donor standardised Trikafta-vs-DMSO separation with raw, size-adjusted and size-band well readouts. Median raw 4.73, adjusted 4.35, band 4.39; adjusted better in 8 of 17 donors. Excluding organoids below 1,069 px (a post-hoc threshold): median 4.73 -> 4.75, better in 9 of 17, Wilcoxon $p = 0.68$. The size effect is real but does not change calls.

4.5 Segmentation benchmark

Table 5. Instance segmentation on the OrgaSegment eval split (12 images).

model	eval mAP@0.5	sd	source file
U-Net v1, 256 px, default post-processing	0.634	0.157	results/seg_eval.json
U-Net v1 + val-tuned post-processing + flip TTA	0.710	0.127	results/seg_eval_tuned_256.json
U-Net v2, 512 px, 60 epochs, default post-processing	0.691	0.135	results/seg_eval_512.json
U-Net v2 + val-tuned post-processing + flip TTA	0.733	0.129	results/seg_eval_tuned_512.json
OrgaSegment (published, Mask R-CNN)	0.760	0.120	Lefferts et al. 2024

Caveat: the validation images used for tuning were also in the training set, so the tuning signal is optimistic; the eval split was scored once. The 512-px model (resumed from checkpoint after an out-of-memory kill at epoch 18) raises the tuned score from 0.710 to 0.733, still below the published mean. With 12 eval images and sd near 0.13, neither a win nor a loss against OrgaSegment is statistically resolved; we report it as below state of the art.

4.6 Organoid-to-tissue fidelity across 151 GEO series

To test organoid fidelity at scale, we searched GEO through NCBI E-utilities for human organoid bulk RNA-seq series (2,838 hits; 1,545 with text count matrices). For 151 series with a usable gene-level matrix and a title naming a single organ, each sample was converted to log counts per million and averaged. The series profile was then correlated with every GTEx v8 tissue median profile (54 tissues).

$$CPM_{gs} = 10^6 x_{gs} / \sum_g x_{gs}, \quad p_g = (1/S) \sum_s \log(1 + CPM_{gs}) \quad (11)$$

$$r_t^{cent} = \frac{\langle p - \mu_p - (m - \mu_m), G_t - m - \text{mean}(G_t - m) \rangle}{(\| \cdot \| \| \cdot \|)}, \quad m_g = \text{mean}_t G_{gt} \quad (12)$$

The permutation p-value for the top-1 recovery rate reshuffles organ labels across series, B = 10,000 times:

$$p_{perm} = (1 + \#\{b : \text{top1}_b \geq \text{top1}_{obs}\}) / (1 + B) \quad (13)$$

Uncertainty on shares uses a series-level percentile bootstrap:

$$CI_{95} = [q_{0.025}, q_{0.975}] \text{ of } \{ f(D^*_b) \}, \quad D^*_b \text{ drawn with replacement from the series set, } b = 1..2000 \quad (14)$$

Table 6. Tissue recovery and fibroblast best-match share. S3000: Spearman on the 3,000 most tissue-variable genes; SALL: all genes; CENT: GTEx-centred Pearson; _NOCULT: 853 MSigDB Hallmark proliferation/MYC/EMT genes removed. Clean subset: 114 series whose summary confirms a single organ and mentions organoids (113 with profiles). Files: results/geo_fidelity_methods.json, results/geo_fidelity_clean.json.

method	top-1 (all)	null top-1	p_perm	top-1 (clean)	fibroblast best (clean)	95% CI
S3000	0.272	0.060	1e-04	0.301	0.230	[0.15, 0.30]
SALL	0.331	0.080	1e-04	0.327	0.257	[0.18, 0.34]
CENT	0.252	0.056	1e-04	0.274	0.363	[0.27, 0.45]
SALL_NOCULT	0.331	0.079	1e-04	0.327	0.248	[0.17, 0.33]
CENT_NOCULT	0.258	0.059	1e-04	0.274	0.425	[0.34, 0.51]
SALL_NONPAR	0.417	0.103	1e-04	0.398	0.204	[0.13, 0.28]
CENT_NONPAR	0.331	0.088	1e-04	0.345	0.372	[0.28, 0.46]
SALL_PEARONLY	0.404	0.114	1e-04	0.389	0.283	[0.20, 0.36]
CENT_PEARONLY	0.377	0.112	1e-04	0.398	0.221	[0.15, 0.29]

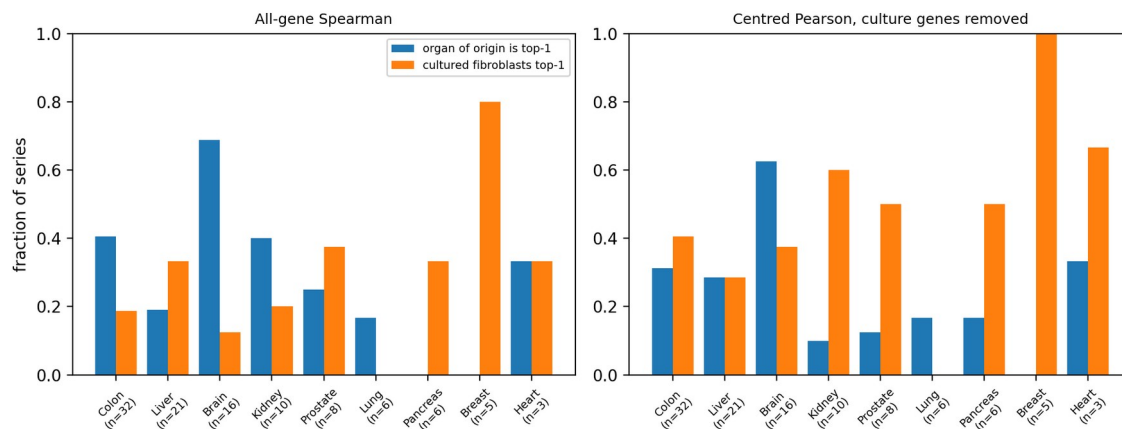


Figure 2. Per-organ share of clean-subset series whose organ of origin (blue) or GTEx cultured fibroblasts (orange) is the top-1 match. Organs with $n \geq 3$.

Organ of origin is recovered four to five times above chance. Brain organoids are the most faithful, and lung, breast and pancreas the least. Across every method, the most frequent best match is GTEx cultured fibroblasts, which we name the culture-fibroblast attractor. It survives the label audit and the removal of proliferation and EMT genes. It is not yet a discovery: GTEx tissues contain immune, vascular and stromal cells, while organoids and cultured fibroblasts are purified cultures, so cell-type purity is an unexcluded explanation. Falsifiable prediction: restricted to epithelium-specific genes, the fibroblast best-match share in the clean subset will stay above 15%. The tool vorganoid fidelity scores any count matrix this way.

4.7 Purity-corrected fidelity (Human Protein Atlas)

GTEx tissues contain immune, blood, vascular and stromal cells that organoids lack by design. Using the Human Protein Atlas single-cell type atlas, we removed 7,595 genes enhanced in any such cell type (NONPAR), or kept only the 9,699 genes enhanced solely in parenchymal types (PARONLY). Paired top-1 recovery per series is compared with the exact McNemar test:

$$p_{McN} = 2 * \sum_{k=0}^{\min(b,c)} C(b+c, k) 0.5^{b+c}, \quad b = \text{base-only hits}, \quad c = \text{variant-only hits} \quad (15)$$

Table 7. Organ-of-origin recovery on 151 series, base method vs gene-restricted variant (results/geo_mcnemar.json).

comparison	top-1 base	top-1 variant	b	c	exact p
SALL vs SALL_NO CULT	0.331	0.331	1	1	1

SALL vs SALL_NO NPAR	0.331	0.417	3	16	0.0044
SALL vs SALL_PA RONLY	0.331	0.404	10	21	0.071
CENT vs CENT_NO CULT	0.252	0.258	0	1	1
CENT vs CENT_NO NPAR	0.252	0.331	4	16	0.012
CENT vs CENT_PA RONLY	0.252	0.377	4	23	0.00031

Purity restriction raises recovery (all-gene Spearman 33% -> 42%, $p = 0.0044$; centred Pearson 25% -> 38%, $p = 0.00031$). Removing culture genes does not change it. Standard whole-transcriptome fidelity scores are therefore biased low for organoids. The prediction stated before this test (fibroblast best-match share above 15% on parenchymal-only genes) held: 28% [0.20, 0.36] and 22% [0.15, 0.29]. The lower bound of the second interval touches the threshold, so purity is weakened but not excluded as the explanation. vorganoid fidelity --purity applies this correction to any count matrix.

5. Negative results (kept by design)

- The initial power-law ($\alpha > 1$) interpretation was withdrawn after large-organoid checks contradicted it.
- Size-adjusted and size-filtered readouts do not improve donor-level modulator discrimination.
- Replication on the public FIS time series is impossible: it is well-level.
- Our segmentation is below the published state of the art.
- Fidelity: 'Pancreas' as best match (35/151 series with S3000) disappears with other methods, so it is a method artefact.
- Fidelity: a 'liver disease' series (GSE278954) ranks Liver 50th of 54, and lung and breast organoids are rarely matched to their organ.
- Fidelity: organ labels come from text, and series profiles average all samples, including any non-organoid controls.

6. Discussion

Small organoids respond less to CFTR modulators, relative to DMSO, than large ones, and the response saturates above a size threshold. Because geometry predicts the reverse, the likeliest explanations are biological: small organoids may lack a mature lumen, contain fewer differentiated secretory cells, or have a closed tight-junction seal later. Falsifiable prediction: in time-lapse single-organoid data from an independent lab, the Trikafta-minus-DMSO effect in the top starting-size quartile will exceed that in the bottom quartile by at least 0.1 log units in most donors. Checking this needs a second single-organoid FIS dataset, which we did not find in public form.

Practical note. Well-level FIS readouts depend on the size mix. Labs comparing drug response across passages or sites should report the starting-size distribution.

Novelty. We did not find this effect reported for intestinal FIS in the literature we searched (OrgaSegment; the FIS_analysis pipeline, which sums area per well). Related: nasal 2D-derived organoids show swelling variation concentrated in large structures (bioRxiv 2021.07.20.453105). This was not a systematic review.

7. Tools used

Table 8. Tools genuinely used (results/tools_ledger.csv): 18 entries, 12 counting toward the gate after excluding infrastructure. Target of 40 not reached.

tool	kind	where used	gate
NumPy	library	src/vorganoid; src/geo_fidelity.py; paper/build_paper.py	counts
SciPy	library	src/geo_fidelity.py (spearmanr); paper/build_paper.py	counts
pandas	library	src/geo_fidelity.py; paper/build_paper.py	counts
scikit-image	library	src/vorganoid (instance post- processing; watershed/labels)	counts
PyTorch	library	train_seg.py; tune_seg.py (U-Net CNN)	counts
Pillow	library	train_seg.py image loading	infra (excluded)

python-docx	library	paper/paperkit.py	infra (excluded)
pytest	test framework	tests/ (hermetic suite)	infra (excluded)
LibreOffice (soffice headless)	application	paper docx -> pdf render	infra (excluded)
poppler pdfinfo	application	page count of rendered paper	infra (excluded)
Git + GitHub	service	repo hosting and pushes	infra (excluded)
NCBI E-utilities (esearch/esummary)	web API	src/ geo_candidates.py	counts
NCBI GEO FTP/HTTPS supplementary files	data service	src/geo_fidelity.py	counts
GTEX v8 gene median TPM (GTEX portal bucket)	reference data service	data/ref/ gtex_v8_median_tpm.gct.gz; src/geo_fidelity.py	counts
MSigDB Hallmark gene sets v2023.2 (Broad GSEA-MSigDB)	gene-set resource	data/ref/ h.all.v2023.2.Hs.symbol.gmt; src/geo_fidelity_methods.py (culture-confound test)	counts
matplotlib	library	src/ fig_geo_fidelity.py; paper figures (fig_size_attenuation.png, fig_geo_fidelity.png)	counts
Human Protein Atlas single-cell type API	database/web API	data/ref/ hpa_single_cell_type_v_api.tsv.gz; src/geo_fidelity_methods.py (purity gene sets)	counts
statsmodels	library	src/ geo_mcnemar.py (exact McNemar tests)	counts

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